

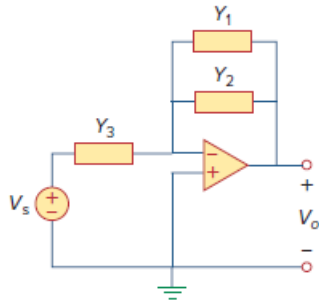
Problem

Realize the transfer function

$$\frac{V_o(s)}{V_s(s)} = -\frac{s}{s+10}$$

using the circuit in Fig. 16.110. Let $Y_1 = sC_1$, $Y_2 = 1/R_1$, $Y_3 = sC_2$. Choose $R_1 = 1 \text{ k}\Omega$ and determine C_1 and C_2 .

Figure 16.110:



Step-by-step solution

Step 1 of 2

We apply KCL at non inverting terminal of Op - Amp

$$(V_s - 0)Y_3 = (0 - V_o)(Y_1 - Y_2)$$

$$Y_3 V_s = -(Y_1 + Y_2) V_o$$

$$\frac{V_o}{V_s} = \frac{-Y_3}{Y_1 + Y_2}$$

$$\text{let } Y_1 = sC_1, \quad Y_2 = \frac{1}{R_1}, \quad Y_3 = sC_2$$

$$\frac{V_o}{V_s} = \frac{-sC_2}{sC_1 + \frac{1}{R_1}} = \frac{\frac{-sC_2}{C_1}}{\frac{s+1}{R_1 C_1}}$$

Step 2 of 2

Comparing this with given transfer functions

$$\frac{C_2}{C_1} = 1, \quad \frac{1}{R_1 C_1} = 10$$

$$\text{If } R_1 = 1 \text{ k}\Omega$$

$$C_1 = C_2 = 100 \mu\text{F}$$